Intergenerational Housing Misallocation and Fertility
Preliminary Note

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July 2026

Abstract

Children require goods and housing space. Family-sized rentals are limited, while access to owner housing requires liquid wealth and borrowing capacity. Older households meanwhile hold much of the large-home stock. I study how this allocation of space changes the private price of children. A compact overlapping-generations model expresses rental limits, down-payment constraints, and old-owner retention as goods-equivalent wedges. It separates changes in family size among incumbent households from changes due to housing-market entry and tenure switching. A finite-life quantitative model adds income risk, saving, tenure and house-size choice, transaction costs, bequests, and children aging out of the household. The current calibration suggests that access to family-sized space is valuable to households near the fertility margin, while the timing of purchases is central to the response of births to housing policy.

Status of this draft

This document is an extended research update rather than a completed paper. The compact model has been checked independently, and the results stated below include the assumptions needed for the derivations. The quantitative sections report a July 9, 2026 calibration snapshot. Two implementation issues are relevant for circulation. Current housing choices must enter demand and moments contemporaneously; this correction changes targeted moments and requires re-estimation. Separately, the production source had drifted from the actual July runtime. The recovered runtime already used the intended entry-age income profile and normalized working-age earnings, so restoring those settings fixes reproducibility rather than changing the July estimates. All counterfactual magnitudes remain provisional until the timing-corrected model is re-estimated.

The quantitative environment below follows the specification used to produce the snapshot: the owner premium applies to usable housing space, bequest utility is not normalized across family-size states, and all owners face the same debt floor. The policy exercises are preliminary patterns from an internally consistent diagnostic runtime. They will be rerun from the corrected and re-estimated baseline.

1 Introduction

A child changes a household's housing problem. It is not only another set of goods to buy, or more time to spend at home. It is often a need for a different dwelling: another bedroom, a larger living space, or a unit where raising a child is feasible. In many high-cost cities this margin is hard to adjust. Large rental units are scarce, family-sized homes are concentrated in ownership, and ownership requires down payments and borrowing capacity at ages when many households have little wealth.

This paper studies whether those space and financial constraints affect final family size and the number of births generated by a cohort. That distinction matters. A housing shock can move births across years without changing completed fertility. It can also change local births by changing who enters a high-cost housing market, even if completed fertility among incumbent households moves little. The question here is when housing constraints change completed fertility, when they mainly change entry and local scale, and how both effects depend on who can access family-sized space over the lifecycle.

The mechanism has three parts. First, children raise the household's demand for space. If the rental market does not offer sufficiently large units, the shadow price of an additional room can exceed the posted rent. Second, the natural substitute is ownership, but buying a family-sized home requires liquidity and credit. A young household may value the larger unit and still be unable to purchase it. Third, older owners may remain in large homes even when their own marginal value of space is low, because transaction costs and tax rules subsidize staying put. Housing scarcity then shows up not only as a high price of space, but also as a mismatch between who occupies family-sized units and who values them for fertility.

A compact two-period model expresses the rental space constraint, the ownership finance constraint, old-owner lock-in, and the fertility distortion as goods-equivalent wedges. I then embed the same economics in a finite-life quantitative model. The model combines lifecycle fertility choice (Sommer, 2016; Doepke and Kindermann, 2019) with housing tenure choice, collateral constraints, and owner-occupied housing (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). The compact housing block follows the one-market floorspace logic in Coven et al. (2025): all housing services clear in one market, while tenure, borrowing, and tax rules determine who can actually occupy family-sized units.

The current quantitative exercise is informative about the mechanism but not yet a final estimate of its size. The July snapshot places substantial fertility-response weight on households constrained by the rental-size limit, and it produces a sharp coincidence of births and purchases near the end of the fertile window. It also misses early ownership, young liquid wealth, and old-age exit from ownership. The first two misses are affected by the implementation corrections described above. I therefore use the current calibration to organize the evidence and the next estimation step, not to make a settled quantitative claim.

2 Related Literature

The paper connects work on fertility choice to work on housing tenure and the allocation of large homes. Since Becker (1960) and Becker and Lewis (1973), economics has treated children as choice variables that enter preferences and require resources. Quantitative versions place that choice in lifecycle environments with income risk, saving, timing, and intrahousehold bargaining (Jones et al., 2010; Sommer, 2016; Doepke and Kindermann, 2019). Recent work emphasizes that the U.S. fertility decline is difficult to explain with one postponement margin or period shock (Kearney et al., 2022).

Housing prices, housing wealth, and mortgage finance also move fertility behavior. Dettling and Kearney (2014) find opposite house-price effects for owners and nonowners, Lovenheim and Mumford (2013) find that housing wealth raises homeowners' fertility, and Cumming and Dettling (2024) show that mortgage-rate pass-through affects births through cash flow. Hacamo (2021) and Dettling and Kearney (2025) connect mortgage access to home purchases and childbearing. In equilibrium, Couillard (2025) show that the supply of large units matters for aggregate fertility.

The outcome here is closer to completed fertility than to a current birth rate. Simon and Tamura (2009) find that rents per room predict lower fertility, including in completed-fertility specifications. Clark (2012) provides an important caution: expensive housing markets delay first births but do not clearly reduce completed fertility. More recent evidence links housing access to both timing and final family size (Japaridze and Sayour, 2024; Dettling and Kearney, 2025; van Doornik et al., 2025). I use this evidence to keep three objects separate: current births, completed fertility among incumbent households, and cohort births after housing-market entry responds.[1]

The closest housing-allocation paper is Coven et al. (2025). Their model studies how property taxes, capitalization, and financial constraints allocate housing between older homeowners and younger constrained households. I add children to that allocation problem. Family-sized space held by an incumbent can then change the private price of children for a young household. The retention force also relates to mortgage lock-in and housing ladders (Fonseca and Liu, 2023; Fonseca et al., 2026). My baseline quantitative model uses transaction costs rather than fixed-rate mortgage lock-in, but the allocation logic is similar.

The quantitative housing block builds on models in which tenure, collateral constraints, and owner-occupied housing shape household behavior (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). It also relates to spatial work in which housing supply and local prices allocate households across markets (Rosen, 1979; Roback, 1982; Gyourko et al., 2013; Hsieh and Moretti, 2019). The present model keeps one housing market so that tenure, family size, and the allocation of space remain the central margins.

3 Analytical Model

3.1 Environment

The economy is populated by overlapping generations of young and old households. A young household has type $i = (y_i, a_i)$: income and wealth at entry. Wealth is inherited from the previous old generation[2]. A household decides whether to enter the economy; inside, she chooses a tenure (rent or own), a house size, completed fertility, and savings for old age. Old households, indexed by j, are of two kinds. Old incumbents, last period's young owners, who enter old age with financial wealth, the house they bought, and a possible accrued gain, choose how much of the house to keep and how large an estate to leave, then exit. The other type is old renters, last period's young renter entrants, who rent in old age and carry no wedge. Nonentrants stay outside the market in both periods. Each period a mass $\bar{M}$ of potential young households enters the pool: the maturing children of the previous cohort plus a renewal inflow. Types are distributed according to an exogenous G_Y.

Preferences. Households have preferences over consumption, housing, and children. Children raise the household's consumption and housing needs: each child claims $\chi > 0$ units of consumption and $\kappa > 0$ units of space, so with consumption spending c_i, housing h_i, and fertility n_i,

$$u_i^Y = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i.$$

[1] Life-course evidence documents that fertility, homeownership, dwelling size, and residential transitions move together (Mulder and Billari, 2010; Kulu and Steele, 2013; Chudnovskaya, 2019). The model treats those joint transitions as economic choices over the lifecycle.

[2] Distributed randomly, without a tie to bequests and inheritants. It can be distributed unequally, however.

Old households value consumption, housing, and the bequest $e_j \geq 0$ they leave at exit:

$$u^O = \log c_j^O + \gamma \log h_j^O + \mathcal{B}(e_j),$$

with $\mathcal{B}$ a warm-glow function, increasing and concave, possibly zero, common to all old households. The exiting cohort's estates are what the arriving cohort inherits.

Housing. There is a fixed stock $\bar{H}$ of divisible floorspace, held by old incumbents and by passive absentee landlords: buyers purchase from either, renters occupy landlord space, and rents and sale proceeds are transfers.[3] Renting and owning are occupancy modes with different size limits, the largest sizes available to rent and to own,

$$h \leq \bar{h}^R \text{ (rent)}, \qquad h \leq \bar{h}^O \text{ (own)}, \qquad \bar{h}^R < \bar{h}^O,$$

Family-sized houses are mostly available to buy, not to rent. All floorspace trades in one market: the asset price is P, and no-arbitrage ties the rental rate to it,

$$q = \rho P, \qquad \rho = r + \delta + \tau^p.$$

The renter pays q as rent; the owner pays it as mortgage interest, property tax, depreciation, and the forgone return on her equity—different payments, equal per-unit value by arbitrage. Given q, a higher property tax lowers P and with it the down payment on any given house.

Financing. A buyer finances at most a share $\phi \in (0, 1)$ of the purchase, so she posts a down payment $(1 - \phi)Ph$ at purchase, out of her liquid wealth. Households can save in an exogenously provided bond, but renters cannot borrow. The interest rate r is exogenous.

Taxes. Two taxes appear: the property tax τ^p, part of the user cost above, and the capital-gains tax. Selling a house triggers a tax: the seller pays the rate τ^g on the difference between the sale price and what she paid, in dollars. Dying instead passes the house with a stepped-up basis - the heir's purchase price is reset to the current value -so the gain is never taxed[4].

Where do gains come from when the real price never moves? From inflation. Dollar prices and incomes all grow at rate ϖ, so relative prices and real choices are unaffected; but the tax form compares dollars from different dates, and a house bought one period ago shows a paper gain even though its real value is unchanged. The tax on that paper gain is real,

$$\bar{\ell} = \tau^g \, \frac{P\,\varpi}{1 + \varpi}$$

per unit sold, and this is the model's only contact with inflation. Every cohort carries the same gain, so every incumbent's unit bears $\bar{\ell}$ if sold while alive.[5] An estate tax, by contrast, would fall on the transfer in any form and leave the sell-or-keep margin undistorted; the gains tax distorts it precisely because death forgives it. Everything is deterministic: a buyer knows the tax her future self will face [6].

[3] There is no construction: the model allocates a fixed stock. This goes together with the stationary equilibrium focus. More sophisticated supply will follow in the extended model and the quantitative application.

[4] This was the essence of the point suggested by Simon. I will go a bit deeper into it in the next weeks, but it is one of the two key margins (the other being financial constraints). The idea is then: agents have an incentive to hold on to the house, because this way neither they nor their heirs pay taxes.

[5] The statutory exclusion, which exempts gains below a dollar threshold, is an institutional detail I suppress; with it, small holdings escape the tax and lock-in sorts by size.

[6] So I am resorting to a trick here to have a capital gain in a stationary model. Coven et al. (2025) have a real growth term there, but they don't have my problem with fertility. More thought is required on this.

3.2 Households

Conditional on entry, household i chooses a tenure mode $m \in \{R, O\}$, housing, fertility, consumption and savings; old age is discounted at β. For $m \in \{R, O\}$,

$$W_i^m = \max_{c_i, h_i, n_i, a_i'} \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i \\ + \beta\big[\mathbf{1}\{m = R\}\, V^R(a_{j'}) + \mathbf{1}\{m = O\}\, V^O(a_{j'}, h_i)\big]$$

subject to

$$c_i + qh_i + a_i' = y_i + a_i, \qquad h_i \leq \bar{h}^m, \qquad a_i' \geq 0,$$
$$\mathbf{1}\{m = O\}\,(1-\phi)Ph_i \leq a_i.$$

Financial wealth carried into old age is $a_{j'} = (1+r)\,a_i' - \mathbf{1}\{m = O\}\, qh_i$. Children have a market price: each child claims χ goods and κ units of space, and space rents at q, so providing for a child costs $\chi + \kappa q$ per period. A useful benchmark is equal scaling, $\chi = \kappa q/\alpha$: the child splits her cost between space and goods in the proportion $\kappa q/\chi = \alpha$, the same proportion in which her parents split their own spending.

The old incumbent enters with financial wealth a_j, housing H_j^0, and an accrued gain; old households have no income. She faces a choice with respect to her housing: sell now and pay the tax, or keep it until death, when the tax is forgiven.[7] She solves

$$V^O(a_j, H_j^0) = \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j)$$
$$\text{subject to} \quad c_j^O + e_j + q\, h_j^O = a_j + qH_j^0 - \bar{\ell}\,(H_j^0 - h_j^O), \qquad 0 \leq h_j^O \leq H_j^0.$$

Selling a unit nets $q - \bar{\ell}$; hence the stepped-up basis acts as a subsidy to locking in. An old renter has no carried housing and no tax:

$$V^R(a_j) = \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \quad \text{subject to} \quad c_j^O + e_j + q\, h_j^O = a_j, \qquad 0 < h_j^O \leq \bar{h}^R,$$

and at an interior choice her marginal value of space is exactly q.

Using $P = q/\rho$, owner housing is bounded by

$$H_i^O = \min\left\{\bar{h}^O,\ \frac{a_i\,\rho}{(1-\phi)\,q}\right\}, \qquad H_i^R = \bar{h}^R.$$

H_i^m is the effective family-housing cap: a renter is capped by the stock, a buyer by the size limit or by collateral, whichever is shorter. Tenure is $W_i^Y = \max\{W_i^R, W_i^O\}$. Entry compares the inside value with an outside option $\bar{W}^E$ under extreme-value shocks of scale κ_E:

$$\pi_i^E = \frac{\exp\{W_i^Y/\kappa_E\}}{\exp\{W_i^Y/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}$$

is the probability that household i enters.

[7] In the more realistic model we will have indivisible housing, so this will look different. With an indivisible house the choice is all-or-nothing: a strict keeper escapes the bill entirely, lock-in concentrates into a notch at the downsizing margin, and pre-saving becomes plan-contingent.

3.3 Equilibrium

Aggregate demand adds young households, old incumbents, and old renters, and the single market clears:

$$\bar{M}\int \pi_i^E\, h_i\, \mathrm{d}G_Y(i) + \int h_j^O\, \mathrm{d}F_O(j) + \int h_j^O\, \mathrm{d}F_R(j) = \bar{H},$$

where F_O and F_R are the measures of old incumbents and old renters. A stationary equilibrium consists of a user cost q (and $P = q/\rho$), young policies and entry probabilities, old policies, and stationary objects $(\bar{M}, G_Y, F_O, F_R)$ such that:

1. the policies solve the household problems above at q;
2. the floorspace market clears, as displayed;
3. the population reproduces itself: young owners become the old incumbents, young renters the old renters, and births plus the renewal inflow recreate the cohort;

3.4 Equilibrium policies

The Lagrangian for a young agent is

$$\begin{aligned}\mathcal{L} = u_i^Y + \beta\, V_{j'} + \lambda_i\big(y_i + a_i - c_i - qh_i - a_i'\big)\\ + \mu_i\big(a_i - (1-\phi)Ph_i\big) + \theta_i\big(\bar{h}^m - h_i\big) + \xi_i\, a_i',\end{aligned}$$

with $\mu_i = 0$ in the rental mode and $\xi_i \geq 0$ the multiplier on the no-borrowing bound $a_i' \geq 0$ for a renter. The formulas below apply at the stated interior saving and retention margins; at a corner, the corresponding complementary-slackness condition replaces the equality.

The first-order condition for saving sets the marginal value of resources today against the return tomorrow:

$$\frac{1}{c_i - \chi n_i} = \beta(1+r)\,\frac{1}{c_{j'}^O} + \xi_i, \qquad \xi_i\, a_i' = 0,$$

which for savers ($\xi_i = 0$), under logarithmic old utility (warm glow $b\log e$ included) and interior old choices, collapses to a savings rule:

$$a_i' = \beta\,(1+\gamma+b)\,(c_i - \chi n_i) + \mathbf{1}\{m = O\}\,\frac{\bar{\ell}\, h_i}{1+r},$$

with γ the old housing weight and b the warm-glow weight: the household saves old age's claim on adult consumption- $k \equiv \beta(1+\gamma+b)$ -plus her discounted future tax bill, so anticipating the lock-in tax is itself a savings motive. By the same conversion, a buyer pays an effective price $q^O = q + \bar{\ell}/(1+r)$ per unit of space—the market price plus her own discounted lock-in tax, pre-funded through savings rather than paid as a flow—while a renter pays q; write q^m for the mode's effective price. A unit of adult consumption commits $1+k$ units of resources, one spent now and k carried.

Then, with no binding constraints, with lifetime resources $w_i = y_i + a_i$,

$$c_i - \chi n_i = \frac{w_i}{1+\alpha+\vartheta+k}, \qquad n_i = \frac{\vartheta\, w_i}{(1+\alpha+\vartheta+k)(\chi+\kappa q^m)}, \qquad h_i = \frac{\alpha\, w_i}{(1+\alpha+\vartheta+k)\, q^m} + \kappa n_i,$$

Dividing the housing derivative of $\mathcal{L}$ by the consumption derivative, $\lambda_i = 1/(c_i - \chi n_i)$, prices every multiplier in goods. For a renter this gives

$$\frac{\alpha\,(c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R, \qquad \zeta_i^R = \frac{\theta_i}{\lambda_i} \geq 0:$$

the left side is u_h/u_c, her marginal rate of substitution between space and consumption—the goods she would give up for one more unit of space, holding utility fixed—and it exceeds the rent exactly by the bite of the rental size limit, computable from her observed bundle. For a young owner the same two derivatives give

$$\frac{\alpha\,(c_i^O - \chi n_i^O)}{h_i^O - \kappa n_i^O} = q^O + \zeta_i^O, \qquad \zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i}(1-\phi)P}_{\zeta_i^{O,F},\ \text{down payment}} + \underbrace{\frac{\theta_i}{\lambda_i}}_{\text{size limit}} :$$

her marginal value of space exceeds her effective price by the bite of whichever access constraints bind—the down payment (μ_i) or the owner size limit (θ_i). When the size limit is slack, the bundle identifies the financing bite alone (F for financing): $\zeta_i^{O,F}$ is the observed ratio minus q^O. For an old incumbent at an interior retention choice,

$$\frac{\gamma c_j^O}{h_j^O} = q - \bar{\ell} :$$

she keeps marginal space she values below what the market would pay, because selling it would realize the taxable gain. Old renters carry no such wedge: their marginal value is q.

The solved policies give the level of fertility; the first-order condition behind them shows where housing enters it. Dividing that condition by the marginal utility of consumption prices the marginal child in goods: in either tenure,

$$\underbrace{\frac{\vartheta\,(c_i^m - \chi n_i^m)}{n_i^m}}_{\text{willingness to pay for a marginal child}} = \underbrace{\chi + \kappa\,(q^m + \zeta_i^m)}_{\text{full price of a child}}.$$

A child's space requirement is priced at the household's own shadow value of space, so a binding access constraint acts as a tax of $\kappa\,\zeta_i^m$ per child. Children get more expensive exactly where family-sized space is hard to reach, with market prices unchanged.

3.5 Efficiency

This section is very preliminary, and serves as an illustration of what one can do with the wedges. The planner faces the economy's real constraints and nothing else: the stock, the size limits, goods feasibility, and the domain conditions. It does not face household budgets, since allocations are supported by lump-sum transfers, and it does not face the financing inequalities, which restrict market purchases rather than occupancy.

Proposition 1. *Hold the stationary cross-section, entry, tenure assignments, and savings fixed; the variation below resizes two owner-occupied houses and converts no tenures. The planner's surplus from moving one unit of space from an old incumbent at an interior retention choice to a young owner whose owner size limit is slack and whose financing constraint binds ($\zeta_i^{O,F} > 0$) is*

$$\underbrace{\left(q^O + \zeta_i^{O,F}\right)}_{\textit{buyer's marginal value}} - \underbrace{\left(q - \bar{\ell}\right)}_{\textit{incumbent's marginal value}} = \zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell} > 0.$$

If this compensated local reassignment is feasible at zero real implementation cost and the displayed surplus is positive, the competitive allocation does not solve the planner's conditional allocation problem.

The buyer's marginal value is her effective price plus her gap: what she would pay for one more unit. The incumbent's is the net-of-tax sale price: what releasing one more unit nets her. In the difference the market price cancels, and the two tax terms that remain are transfers the planner recovers, not resource costs. If the move must be implemented through costly financing instruments rather than direct reassignment, their marginal real cost subtracts from this surplus; if the down-payment constraint is a primitive nothing relaxes, the gap is not recoverable. The long version carries this implementation cost explicitly.

Figure 1 draws a solved stationary case as theory. The buyer's marginal value of space falls along her constrained path and sits above her effective price by the financing gap $\zeta^{O,F}$ at her cap, while the incumbent's sits below the market price by her tax $\bar{\ell}$. Pay the incumbent for one unit at her value, $q - \bar{\ell}$; charge the buyer at hers, $q^O + \zeta^{O,F}$: both are exactly as well off as before, and $\zeta^{O,F} + \bar{\ell}/(1+r) + \bar{\ell}$ in goods is left over.

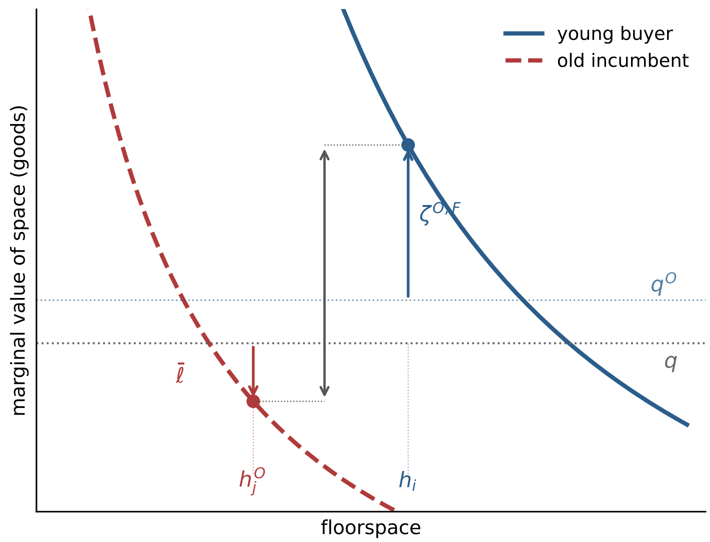

Figure 1: Misallocation. The collateral-capped buyer values marginal space at $q^O + \zeta^{O,F}$, above the market price; the locked-in incumbent values it at $q - \bar{\ell}$, below; the distance between the two points is the recoverable surplus.

3.6 Fertility and policy

Fix a tenure mode and a household whose cap binds, so that her housing is pinned at the effective family-housing cap defined in the household problem: the rental size limit for a renter, the shorter of the owner size limit and the collateral bound for a buyer. Write H for the level of that cap, $w = y + a$ for lifetime resources, and q^m for the mode's effective price. We relax the cap by a unit of space and study fertility. Concentrating the savings rule into the budget, fertility at the cap solves $\vartheta/n = (1+k)\chi/(w - q^m H - \chi n) + \alpha\kappa/(H - \kappa n)$; differentiating with respect to the cap level,

$$\frac{\mathrm{d}n}{\mathrm{d}H} = \frac{\dfrac{\alpha\kappa}{(H-\kappa n)^2} - \dfrac{(1+k)\,\chi q^m}{(w - q^m H - \chi n)^2}}{\dfrac{\vartheta}{n^2} + \dfrac{(1+k)\,\chi^2}{(w - q^m H - \chi n)^2} + \dfrac{\alpha\kappa^2}{(H-\kappa n)^2}},$$

which is positive for every binding cap if and only if $\chi \leq (1+k)\,\kappa q^m/\alpha$: relaxing access raises fertility whenever the child's consumption requirement does not exceed its space requirement priced at the lifetime cost of adult consumption. The derivative races two forces: extra space makes the child's space floor κ easier to meet, pushing fertility up, while paying q^m for the space tightens the budget around the child's goods floor χ, pushing fertility down. The savings margin weakens the second force: the housing bill is paid partly by saving less, so only a fraction $1/(1+k)$ of it falls on current spending, and the threshold below which access raises fertility scales up by $(1+k)$. Figure 2 draws the relation: fertility rises along the binding cap at rate $\mathrm{d}n/\mathrm{d}H$ and flattens at h^u, where the cap stops binding.

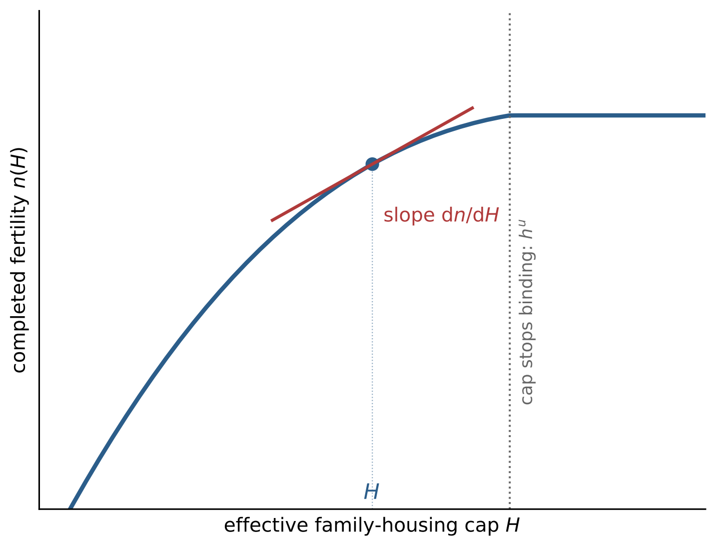

Figure 2: The fertility margin. Completed fertility against the effective family-housing cap, from the same solved case as Figure 1; the slope at the equilibrium cap is $\mathrm{d}n/\mathrm{d}H$, and the cap stops binding at h^u.

One can decompose the effect of a (fertility-neutral) policy on fertility. Let $\bar{n}_i^E$ denote the number of children in the outside option; the baseline normalization is $\bar{n}_i^E = 0$.[8] The access channel runs through the cap, and the cap says where a policy can be effective: a down-payment grant raises a_i, financial deregulation raises ϕ, and either loosens the collateral bound $a_i\rho/((1-\phi)q)$; making family-sized units available to rent raises $\bar{h}^R$. None of this is construction: the stock $\bar{H}$ is unchanged, and the policy changes who can occupy how much of it. At fixed prices, a small policy z then moves aggregate fertility through the constrained:

$$\left.\frac{\mathrm{d}N}{\mathrm{d}z}\right|_q = \bar{M}\sum_m \int_{\mathcal{E}_m(z)} \left[\pi_i^E\,\Psi_i^m + (n_i^m - \bar{n}_i^E)\,\frac{\pi_i^E(1-\pi_i^E)}{\kappa_E}\,\Theta_i^m\right]\mathcal{P}_i^m(z)\,\mathrm{d}G_Y(i),$$

where $\mathcal{E}_m(z)$ is the set of households whose cap binds in mode m, $\mathcal{P}_i^m = \partial H_i^m/\partial z$ is the pass-through of the policy to the binding component of the cap, $\Psi_i^m = \mathrm{d}n/\mathrm{d}H$ is the fertility response, and $\Theta_i^m = \zeta_i^m/(c_i^m - \chi n_i^m)$ drives the entry term. A policy aimed at a slack constraint moves nothing: a down-payment grant does nothing for a household pinned by the rental size limit. At

[8] That is, I am not imposing here a planner solving for an explicit desired level of fertility and studying the housing channel to achieve it. Here, I am just looking at a planner who relaxes housing contraint (to ease the misallocation described above) and observing that that can have a fertility impact.

$\chi = (1+k)\kappa q^m/\alpha$, equal scaling at lifetime prices, Ψ_i^m is proportional to $\zeta_i^m/(c_i^m - \chi n_i^m)$, so the same gap that measures misallocation predicts whose fertility responds.

Because tenure is a discrete choice, a policy that shifts $W_i^O - W_i^R$ also moves households across the rent–own boundary. The gains tax makes that crossing non-trivial. This is also explored in the forthcoming extended version.

4 Quantitative Lifecycle Model

4.1 Lifecycle and Income

The quantitative model embeds the same mechanism in a finite-life economy. It has the standard lifecycle structure: households age, face income risk, save, choose whether to rent or own, choose housing size, decide fertility, and eventually leave bequests. The housing market has one aggregate user cost and one asset price. Tenure matters because it changes the size menu, financing constraints, transaction costs, and tax exposure.

Time is discrete. Households enter adult life at age $a = 1$, retire after age J_R, and die at the end of age J. Before retirement, earnings depend on age and idiosyncratic productivity $z \in \mathcal{Z}$:

$$y(a,z) = (1-\tau_{\text{pay}}) w e_a z, \qquad z' \sim \Pi^z(\cdot|z), \qquad a \le J_R, \tag{1}$$

and retired households receive pension income p^{ret}. The wage is common apart from age and productivity because the quantitative mechanism is housing access, tenure, and lifecycle wealth.

4.2 Preferences and Bequests

Let $m(n,s)$ be the number of dependent children living at home. Encoded completed family size n and current dependents differ because children eventually mature. Household needs are

$$\begin{aligned} \bar{c}(n,s) &= \bar{c}_0 + \bar{c}_n m(n,s), \\ \bar{h}(n,s) &= \bar{h}_0 + \bar{h}_{\text{jump}} \mathbf{1}\{m(n,s) \ge 1\} + \bar{h}_n m(n,s). \end{aligned} \tag{2}$$

The first child can create a discrete housing-space jump; additional dependent children raise the required space linearly.

Flow utility is Stone–Geary over consumption and usable housing space, wrapped in CRRA. The owner service premium applies after household needs are removed:

$$u(c,h,d;n,s) = \frac{\left[(c-\bar{c}(n,s))^{\alpha_c} \mathfrak{h}(h,d;n,s)^{1-\alpha_c}\right]^{1-\sigma}}{1-\sigma} + \psi_{\text{child}} m(n,s), \tag{3}$$

where

$$\mathfrak{h}(h,d;n,s) = \begin{cases} h^R - \bar{h}(n,s), & d = R, \\ \chi_O\left[H_k - \bar{h}(n,s)\right], & d = O_k, \end{cases} \qquad \chi_O > 0. \tag{4}$$

Utility is $-\infty$ when residual consumption or usable space is nonpositive. In market clearing, an owner demands the physical quantity H_k, not the preference-scaled quantity $\chi_O[H_k - \bar{h}(n,s)]$.

At death, households receive warm-glow bequest utility. The house passes to the estate at market value: there is no forced sale, no transaction cost, and no tax at death, so bequeathable wealth is $W^B = b + PH(d)$, liquid wealth plus the gross value of any house held. Bequest utility is

$$\mathcal{B}(W^B, n) = \theta_0(1 + \theta_n n)\frac{(\theta_1 + \max\{W^B, 0\})^{1-\sigma}}{1-\sigma}. \tag{5}$$

This utility is not normalized to zero at a zero estate. Because its scale depends on n, that level is part of the family-size choice. The parameter θ_n lets completed fertility affect the desired estate even after children have left the household.

4.3 Housing, Finance, and Taxes

There is one aggregate housing-services market. The user cost q and asset price P satisfy

$$q = (r + \delta + \tau^p)P. \tag{6}$$

Housing supply is upward sloping:

$$H^S(q) = H_0 \left(\frac{q}{\bar{q}}\right)^{\eta}. \tag{7}$$

The fixed-stock compact model is the limiting case $\eta = 0$ around the benchmark stock.

Households rent or own one of K discrete housing sizes. Write

$$\mathcal{D} = \{R, O_1, \ldots, O_K\}, \tag{8}$$

where R denotes renting and O_k denotes ownership of a house with physical size H_k, ordered so that $H_1 < \cdots < H_K$. Renters choose housing continuously up to $\bar{h}^R$; owners choose from the discrete ladder. The ladder is the quantitative version of the family-housing segmentation in Section 3.

Renters choose c, h^R, b' subject to

$$c + qh^R + b' = Rb + y(a, z), \qquad 0 \le h^R \le \bar{h}^R, \qquad b' \ge 0, \tag{9}$$

where $R = 1 + r$. Owners of rung H_k choose c, b' subject to the owner branch budget and the borrowing limit described below. Owners pay maintenance and property taxes through $(\delta + \tau^p)PH_k$.

At origination, a buyer can finance at most a share ϕ of house value, so the required down payment is

$$(1 - \phi)PH_k. \tag{10}$$

Equivalently, adjusted liquid wealth immediately after purchase must satisfy $\tilde{b} \ge -\phi PH_k$. End-of-period saving for every owner, whether a buyer or an incumbent, faces the same debt floor:

$$b' \ge -\phi PH_k. \tag{11}$$

Housing adjustment is costly. A sale of house H_k incurs the proportional transaction cost ψ^s, so net sale proceeds are

$$\Omega(H_k) = (1 - \psi^s)PH_k. \tag{12}$$

Transaction costs are the only disposal friction in the computed benchmark: an owner who sells pays ψ^s per unit of value and faces no tax on accrued gains.

The capital-gains treatment of Section 3 is specified as an extension and is not part of the computed benchmark. Pricing realized gains requires tracking a tax basis B^p for each owner; a sale would then trigger $\mathcal{T}^g(PH_k, B^p) = \tau^g \max\{PH_k - B^p - \bar{g}, 0\}$ with statutory exclusion $\bar{g}$, net proceeds would fall to $\Omega(H_k) - \mathcal{T}^g(PH_k, B^p)$, and stepped-up basis at death would forgive the accrued gain. The benchmark does not track B^p: the old-owner lock-in wedge $\bar{\ell}$ is priced analytically in Section 3, and the counterfactual section instead bounds its quantitative content with a direct retention-margin experiment.

4.4 Fertility and Child Aging

Fertility is chosen during the fertile window

$$a \in \{A_f^{\text{start}}, \ldots, A_f^{\text{end}}\}.$$

A childless household chooses

$$n' \in \mathcal{N} = \{0, 1, \ldots, \bar{n}\}. \tag{13}$$

The option $n' = 0$ means no birth in the current period. A positive choice fixes completed fertility forever and moves its children into the first dependent stage. A household that reaches the end of the fertile window without a positive choice remains childless.

In the computation, $\mathcal{N} = \{0, 1, 2\}$ and the last element is the open-ended highest family-size category. The completed-fertility moment matched to the CPS is reported as $2\mathbb{E}[n]$. This household-to-woman conversion is applied only to that reported moment. Child needs and bequest utility use the encoded family-size category n, while the housing-size target maps $n = 1$ to the 1–2 group and $n = 2$ to the 3+ group. The distinction is a unit convention and does not create an additional fertility choice.

Let $s = 0$ denote no dependent children and let $s = 1, \ldots, S_c$ denote dependent-child stages. After a positive fertility choice, the household enters $s = 1$. Children age stochastically through the stages according to Π^c. While children are at home, $m(n, s) = n$; after maturity, $m(n, s) = 0$. Completed fertility still matters through bequest utility.

4.5 Household Problem

Within the period, the household first faces fertility if eligible, then chooses tenure and housing position, then chooses consumption, rental housing, and saving. Let the continuation value after income risk and child aging be

$$\bar{V}(b', z, d', a+1, n, s) = \sum_{z'} \sum_{s'} \Pi^z(z'|z) \Pi^c(s'|s, n) V(b', z', d', a+1, n, s'). \tag{14}$$

The terminal value is

$$V(b, z, d, J+1, n, s) = \mathcal{B}(W^B, n).$$

Conditional on renting, the branch value is

$$\begin{aligned} V^R(b, z, R, a, n, s) = \max_{c, h^R, b'} \quad & u(c, h^R, R; n, s) + \beta \bar{V}(b', z, R, a+1, n, s) \\ \text{s.t.} \quad & c + q h^R + b' = Rb + y(a, z), \\ & 0 \le h^R \le \bar{h}^R, \qquad b' \ge 0. \end{aligned} \tag{15}$$

Conditional on owning rung H_k after any adjustment, the branch value is

$$\begin{aligned} V^{O_k}(\tilde{b}, z, O_k, a, n, s) = \max_{c,b'} \quad & u(c, H_k, O_k; n, s) + \beta\bar{V}(b', z, O_k, a+1, n, s) \\ \text{s.t.} \quad & c + (\delta + \tau^p)PH_k + b' = R\tilde{b} + y(a, z), \\ & b' \geq -\phi PH_k. \end{aligned} \tag{16}$$

The debt floor is the same for a buyer and an incumbent. Purchase feasibility is screened separately through the down-payment condition before this branch is evaluated.

Let $H(R) = 0$ and $H(O_k) = H_k$. The liquid wealth available after housing adjustment is

$$\tilde{b}(b, d, d') = b + \mathbf{1}\{d \in \mathcal{O},\ d' \neq d\}\Omega(H(d)) - \mathbf{1}\{d' \in \mathcal{O},\ d' \neq d\}PH(d'). \tag{17}$$

A seller collects net proceeds, a buyer pays the full purchase price, and a household that keeps its position pays nothing.

The tenure branch value for feasible d' is

$$\widetilde{V}^T(d'|b, z, d, a, n, s) = \begin{cases} V^R(\tilde{b}(b, d, R), z, R, a, n, s), & d' = R, \\ V^{O_k}(\tilde{b}(b, d, O_k), z, O_k, a, n, s), & d' = O_k. \end{cases} \tag{18}$$

An infeasible branch is assigned value $-\infty$. Tenure and owner-rung choices have Type-I extreme-value shocks with scale κ_T:

$$\pi^T(d'|b, z, d, a, n, s) = \frac{\exp\{\widetilde{V}^T(d'|b, z, d, a, n, s)/\kappa_T\}}{\sum_{\hat{d}\in\mathcal{D}} \exp\{\widetilde{V}^T(\hat{d}|b, z, d, a, n, s)/\kappa_T\}}, \tag{19}$$

and the inclusive tenure value is

$$V^T(b, z, d, a, n, s) = \kappa_T \log \sum_{\hat{d}\in\mathcal{D}} \exp\{\widetilde{V}^T(\hat{d}|b, z, d, a, n, s)/\kappa_T\}. \tag{20}$$

At the estimated value $\kappa_T = 0$, these expressions mean their deterministic limit: the branch with the highest value is chosen with probability one, and a tie is resolved by the fixed ordering of feasible branches.

For a childless household in the fertile window, the value of fertility option n' is

$$\widetilde{V}^F(n'|b, z, d, a, 0, 0) = V^T(b, z, d, a, n', s(n')), \qquad s(n') = \mathbf{1}\{n' > 0\}. \tag{21}$$

With Type-I extreme-value shocks of scale κ_F,

$$\pi^F(n'|b, z, d, a, 0, 0) = \frac{\exp\{\widetilde{V}^F(n'|b, z, d, a, 0, 0)/\kappa_F\}}{\sum_{\hat{n}\in\mathcal{N}} \exp\{\widetilde{V}^F(\hat{n}|b, z, d, a, 0, 0)/\kappa_F\}}. \tag{22}$$

The value before the fertility draw is the corresponding inclusive value:

$$V(b, z, d, a, 0, 0) = \kappa_F \log \sum_{\hat{n}\in\mathcal{N}} \exp\{\widetilde{V}^F(\hat{n}|b, z, d, a, 0, 0)/\kappa_F\}. \tag{23}$$

Outside the fertile window, or after a positive fertility choice, this stage is degenerate and $V(b, z, d, a, n, s) = V^T(b, z, d, a, n, s)$.

4.6 Stationary Equilibrium

Let g denote the stationary composition of adult households over liquid wealth, productivity, housing position, age, family size, and child age, (b, z, d, a, n, s), normalized to integrate to one. Total adult scale is S, so the corresponding level measure is $G = Sg$:

$$\int dg(b, z, d, a, n, s) = 1, \qquad \int dG(b, z, d, a, n, s) = S. \tag{24}$$

The distinction between g and G separates composition from scale: household policies determine the composition of the population, while entry and replacement determine how many adults the modeled market contains.

Definition 1 (Stationary equilibrium)**.** *A stationary equilibrium is a user cost and asset price (q, P), value functions and policies, tenure and fertility probabilities, stationary composition g, adult scale S, level measure $G = Sg$, and aggregate housing quantities such that:*

1. *households solve the branch problems (15)–(16), tenure choice is given by (19), fertility choice is given by (22), and terminal utility is (5);*
2. *current choices, saving, income risk, child aging, adult aging, death, and the injection of new entrants induce a normalized transition operator $\mathcal{T}(g; q, S)$. Stationarity requires*

$$g = \mathcal{T}(g; q, S); \tag{25}$$

3. *per-unit physical housing demand evaluates each household's current tenure and housing choices:*

$$\begin{aligned}\widehat{H}^D(g, q) = \int \Bigg[&\Pi^T(R \mid b, z, d, a, n, s)\widehat{h}^R(b, z, d, a, n, s) \\ &+ \sum_{k=1}^{K} \Pi^T(O_k \mid b, z, d, a, n, s) H_k\Bigg] \mathrm{d}g(b, z, d, a, n, s).\end{aligned} \tag{26}$$

   where $\Pi^T(d' \mid b, z, d, a, n, s)$ is the current tenure probability and $\widehat{h}^R(b, z, d, a, n, s)$ is expected rental space conditional on renting, both after integrating over the nested family-size choice;

4. *let $E_0(g, q)$ be the inflow of young adults required to reproduce one unit of the stationary composition, $B_0(g, q)$ the mature locally born flow per unit of adult scale, M the outside-origin entrant pool, and $q^E(q)$ the probability that a potential entrant enters the modeled market. Entry balances adult replacement when*

$$SE_0(g, q) = q^E(q)\left[M + SB_0(g, q)\right]. \tag{27}$$

   Provided $E_0(g, q) > q^E(q)B_0(g, q)$, this condition gives the finite scale

$$S = \frac{q^E(q)M}{E_0(g, q) - q^E(q)B_0(g, q)}; \tag{28}$$

5. *the housing market clears in levels and the user-cost relation holds:*

$$S\widehat{H}^D(g, q) = H^S(q), \qquad q = (r + \delta + \tau^p)P; \tag{29}$$

6. *taxes, pensions, and any rebates satisfy the government accounting rule specified for the calibration or counterfactual.*

The benchmark sets $S = 1$ as a normalization; in counterfactuals, S is an endogenous equilibrium object. Computationally, for a candidate q we solve the household problems and the stationary composition, recover S from the entry condition, and update q until level housing demand equals supply. This nesting is a solution method, not a different equilibrium concept.

4.7 Computation

One model period is four years. Period $j = 0, \ldots, 16$ begins at age $18 + 4j$; the last interval covers ages 82–85. Retirement begins at 66, and the last period in which a birth can occur begins at 42. Annual objects are converted before solution:

$$R = (1 + r_{\text{ann}})^4, \quad \beta = \beta_{\text{ann}}^4, \quad \delta = 1 - (1 - \delta_{\text{ann}})^4, \quad \tau^p = 4\tau_{\text{ann}}^p.$$

Income and consumption-flow quantities are expressed in four-year units; housing remains in rooms.

The search grid has $N_b = 120$ nodes on $[-12, 30]$, with a dense core on $[-5, 7]$ and a thinner upper tail beyond 15. Income takes values $z \in \{0.6, 0.8, 1.0, 1.2, 1.4\}$ with invariant weights $\pi_z = (0.1, 0.2, 0.4, 0.2, 0.1)$ and transition matrix $\Pi^z = 0.85I + 0.15\mathbf{1}\pi_z'$. One dependent-child stage has an expected duration of 18 years, giving a four-year maturity probability of 2/9. Entrants have mass $1/J$ each period. The balanced pension equals payroll-tax revenue per retiree in the stationary age distribution.

Value functions are computed by backward induction. Golden-section search solves saving in each branch, and Howard iteration accelerates evaluation. The stationary distribution follows by forward iteration, while damped fixed-point iteration and a scalar bracketing refinement solve the housing market. The intended market-residual tolerance is 10^{-4}; strict rejection of candidates above that tolerance is part of the corrected production run.

Because tenure and fertility choices are close to deterministic at the calibrated point, moments built on buy thresholds and the borrowing constraint move with the wealth grid, and grid resolution matters more here than in a standard lifecycle model. An earlier calibration at $N_b = 60$ re-evaluated 18 percent worse at $N_b = 120$ at the same parameters; part of its fit was grid noise. The July snapshot was searched at $N_b = 120$ and re-evaluated at $N_b = 240$. The corrected estimation will use the same search and verification grids.

5 Calibration

5.1 Externally set objects

The model period is four years. I set standard financial parameters, the income process, and the housing menu outside the estimation. The entrant wealth distribution is taken directly from the PSID.

Table 1: Externally set parameters

Object	Value	Source or rule
Interest rate r	4% per year	standard
Depreciation δ	2% per year	standard
Property tax τ^p	1% per year	U.S. average effective rate
Financed share ϕ	0.80	20% down payment
Sale cost ψ^s	0.06	transaction-cost evidence
Risk aversion σ	2	standard
Bequest shift θ_1	0.01	near-linear warm glow
Payroll tax τ_{pay}	0.179	pension balances in the stationary distribution
Income profile e_a	0.65 at 18, rises to 1.0 at 34, declines after 58	normalized lifecycle profile; five-state persistent income process
Owner menu $\{H_k\}$	$\{2, 4, 6, 8, 10\}$ rooms	room-count support, ACS
Renter cap $\bar{h}^R$	6.0 rooms	90th-percentile rental size rule of Greaney et al. (2025), measured in the matched ACS sample; 6.5 as robustness
Supply (H_0, η)	$(4, 1)$	unit-elastic benchmark
Entry wealth	PSID young-renter wealth-to-income distribution	first stage

The renter cap is the quantitative counterpart of $\bar{h}^R$ in the analytical model. I measure it from the rental-size distribution rather than estimate it internally. The fit is nearly flat between 6.0 and 6.5 rooms and deteriorates outside that range. The owner ladder is likewise measured from the support of rooms in the ACS. These objects therefore describe the available housing products; the estimation determines how households value and sort across them.

5.2 Estimated parameters

I estimate the remaining 13 parameters by simulated method of moments. The objective combines moments on family size, housing consumption, tenure, and lifecycle wealth. I first search globally using differential evolution and then refine the best candidates with local polish chains, using $N_b = 120$ asset-grid points. With fourteen moments, the system has one overidentifying restriction by count, although the local sensitivity matrix is poorly conditioned in several parameter blocks.

Table 2: Internally estimated parameters (July 9 snapshot)

Parameter	Interpretation	Estimate
β_{annual}	Annual discount factor	0.940
α_c	Consumption weight	0.620
$\bar{c}_0$	Baseline consumption requirement	1.279
$\bar{c}_n$	Consumption cost per dependent child	0.116
$\bar{h}_0$	Baseline housing requirement	1.000
$\bar{h}_{\text{jump}}$	Discrete housing-need jump at the first child	2.120
$\bar{h}_n$	Housing requirement per dependent child	1.382
ψ_{child}	Utility from children	0.349
κ_F	Family-size choice scale	1.019
κ_T	Tenure-choice scale	0.000
χ_O	Owner housing-service multiplier	1.150
θ_0	Bequest strength	0.0015
θ_n	Bequest shifter by number of children	0.972

The parameters are disciplined jointly rather than by a one-to-one assignment of moments to parameters. The main sources of variation are as follows.

The *savings block* is disciplined by liquid wealth among young renters, median nonhousing wealth at ages 65–75, and ownership at ages 25–34 and 65–75. The discount factor β governs asset accumulation over the lifecycle. The baseline consumption requirement $\bar{c}_0$ changes the resources available for saving, housing, and children. It is therefore identified only indirectly from the joint wealth and tenure profile, and is more weakly separated from β than the table alone suggests.

The *housing-needs block* uses the level and change in rooms across family states. Mean renter rooms among childless households primarily discipline $\bar{h}_0$; the housing response at the first birth disciplines $\bar{h}_{\text{jump}}$; and the room difference between households with three or more children and those with one or two children disciplines $\bar{h}_n$. These moments also move with income, wealth, and tenure, so the three housing requirements are jointly identified. In local sensitivity calculations their effects are strongly correlated.

The consumption weight α_c and the owner-service multiplier χ_O jointly determine housing demand and the relative appeal of ownership. Their main identifying moments are the owner–renter room gap, the share of owners in homes with at least six rooms, prime-age and young ownership, and the family ownership gap. The room-distribution moments help distinguish a taste for housing services from a taste for ownership, but the separation remains local because finance and lifecycle wealth shift the same tenure margins.

The *family-size block* is disciplined by completed fertility and childlessness. The per-child consumption cost $\bar{c}_n$, the utility shifter ψ_{child}, and the choice scale κ_F all affect these two moments and are not cleanly separated by them. Housing responses across family states provide additional information because $\bar{c}_n$ changes the resources available for housing, but local sensitivity calculations still find this block highly collinear. The tenure scale κ_T is disciplined by ownership rates by age and family status. Its estimate of zero says that the targeted tenure moments do not call for additional taste smoothing; it should not be read as precise interior identification.

Finally, the bequest strength θ_0 is informed by old-age nonhousing wealth and ownership, while θ_n is informed by the parent–childless gap in old-age nonhousing wealth. These are joint restrictions: old-age ownership also responds strongly to β, α_c, and χ_O. Because θ_0 is close to zero, the data have little local leverage over θ_n. An earlier Jacobian audit of a closely related target system was formally full rank at its preferred point but badly conditioned. I therefore treat the estimates as a provisional parameterization, not as evidence that each parameter is tightly identified.[9]

5.3 Target moments and fit

The calibration targets fourteen moments: completed fertility and childlessness from the CPS (the June 2024 supplement cells for women 40–44), tenure and housing-size moments by age and family status from a matched ACS metropolitan sample, and wealth moments from the PSID. Table 3 compares the data with the model at the July 9 parameterization. The weighted objective is 6.31.

Table 3: July 9 target moments and model fit ($N_b = 120$; provisional loss 6.31)

Moment	Data	Model
Old-age ownership rate (65–75)	0.764	0.893
Ownership rate (25–34)	0.341	0.226
Owner–renter mean rooms (no children at home, 30–55)	2.419	2.151
Family ownership gap	0.168	0.298
Young renter liquid wealth / income (25–35)	0.179	0.353
Ownership rate (30–55)	0.575	0.531
Renter mean rooms (no children at home, 30–55)	3.805	3.655
Childlessness rate	0.188	0.260
Rooms gap, 3+ vs. 1–2 children (30–55)	0.368	0.269
Old parent–childless nonhousing wealth gap	1.007	0.842
Housing increment at first birth (rooms)	0.664	0.618
Completed fertility	1.918	1.949
Old nonhousing wealth / income, median (65–75)	2.230	2.328
Owner share with rooms ≥ 6 (no children at home)	0.596	0.593

The July snapshot is close to the fertility level and several size-distribution moments. Old-age ownership is too high: the model keeps 89 percent of 65–75 households in owned housing against 76 percent in the data, because nothing in the model pushes an old owner out of her house. There are no health shocks and no mortality-linked downsizing, so retention is close to absorbing. Ownership at 25–34 is too low (0.23 against 0.34), while young renter wealth is too high (0.35 against 0.18). Both moments are linked to the model's saving and housing-access margins and must be interpreted jointly. Childlessness also overshoots (0.26 against 0.19), even though average completed fertility is close to the data. That combination points to a miss in the distribution of family size rather than its mean.

This table records the pre-repair July snapshot. The correction to the timing of current tenure and housing choices changes several tenure and housing moments, including the same-period housing response at the first birth. I therefore use the table to describe the provisional fit, not as the baseline for the policy results.

Re-evaluated at $N_b = 240$ at the same parameters, the loss rises from 6.31 to 6.84, and the drift concentrates in the family ownership gap and the old-age ownership rate, both of which move further from their targets at the finer grid. This comparison records the numerical drift in the July

[9] Several estimates are at or near the limits of the July search region, including β, $\bar{c}_0$, $\bar{h}_0$, ψ_{child}, κ_F, κ_T, χ_O, and θ_0. The estimates and fit will be updated after the measurement corrections and re-estimation.

snapshot. The corrected calibration will report the same 120/240 comparison.

6 The July 9 Equilibrium Snapshot

The lifecycle pattern is descriptive of the provisional snapshot. In particular, the plotted ownership stock uses the inherited tenure state and therefore records current purchases one period late. The corrected run will replace this figure rather than reinterpret it.

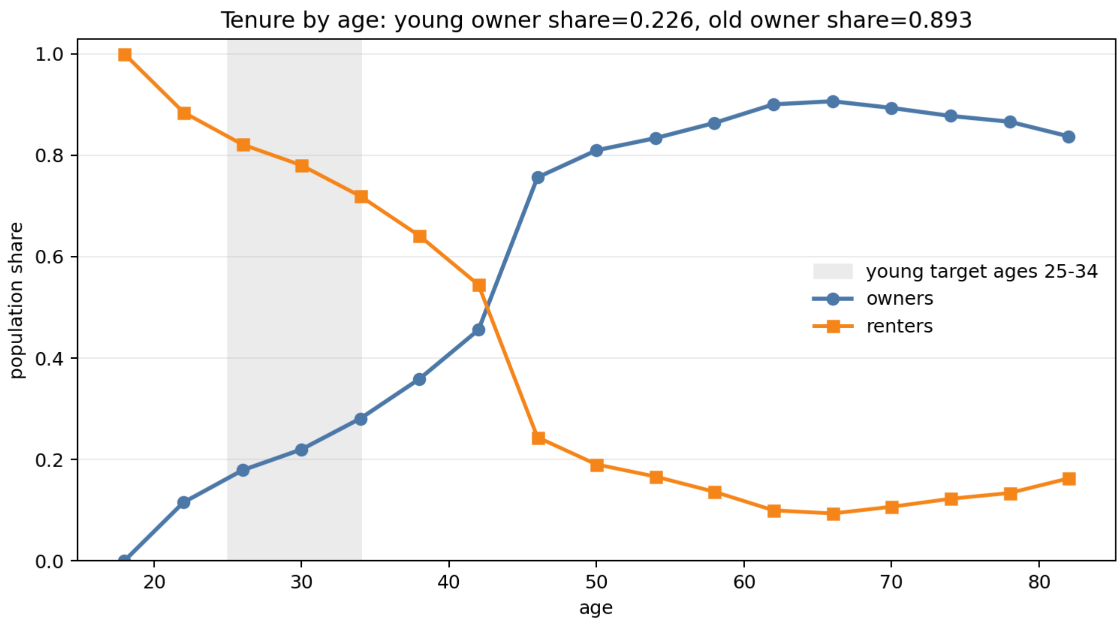

Figure 3: Ownership over the lifecycle in the July 9 snapshot. The steep rise near the end of the fertile window and high old-age ownership are visible, but current purchases are recorded one model period late in this version.

Renting households with children sit against the rental cap, while owners use the discrete room ladder. Family-sized space is scarce through availability: within a size cell, the model rent carries no separate premium. The snapshot also places many young renters near zero liquid wealth. These features remain useful diagnostics, but their age profiles and aggregate weights must be regenerated after the timing correction.

7 The Mechanism Quantified

Section 3's argument runs through one object, the access gap ζ: the amount by which a constrained household's marginal valuation of family space exceeds its market user cost. Under equal scaling, the appendix result says the same gap prices misallocation, fertility exposure, and entry. The July snapshot gives a first measurement of that gap.

The measured gap is large. Among fertile-age renting households pinned at the rental cap, the mean access gap is 0.248 goods units, 1.28 times the market user cost, so capped renters value a marginal room at 2.3 times what the market charges for it. Weighting fertility responses by exposure as in the fixed-price decomposition of Section 3, 73 percent of the model's fertility-response mass sits on households with $\zeta > 0$. Within the July diagnostics, the share is similar across grid resolutions and owner menus. Its level remains provisional because the exposure weights will change when current housing choices are measured contemporaneously.

The model places many purchases near the fertility deadline. The purchase region expands in the last fertile period, when the option value of waiting to have children expires. Buyers in this region hold wealth above the required down payment. In the current model, housing access therefore does not operate only by forcing households to save toward a distant threshold. It also compresses births and purchases into a joint decision near the deadline. The corrected run will determine how much of this pattern survives contemporaneous measurement.

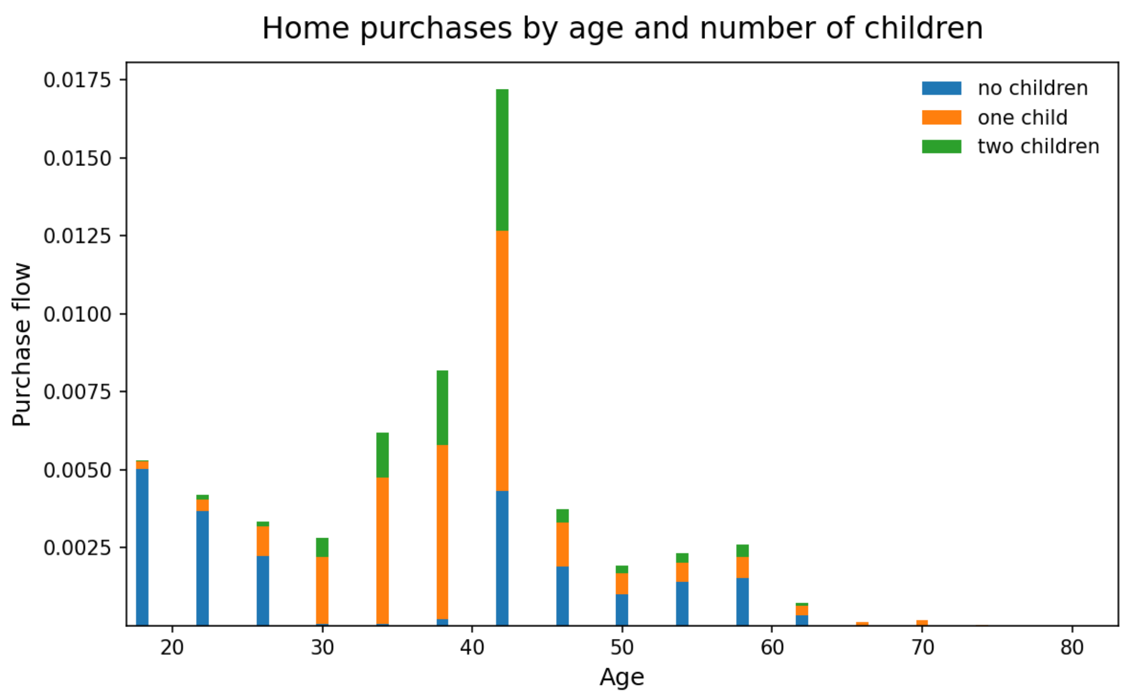

Figure 4: Purchase flows by age and family size in the July 9 snapshot. Buying is concentrated near the end of the fertile window and is birth-linked.

The compact model's misallocation proposition is an accounting statement about who holds family-capable space, and it can be checked against the data. The table reports the allocation of the owner stock with six or more rooms, in the calibrated equilibrium and in the matched ACS sample.

Table 4: Who holds the family-sized owner stock (rooms ≥ 6)

Holder	Data (ACS)	Model
Age 18–29	3.4%	0.4%
Age 30–45	24.8%	12.7%
Age 46–64	43.3%	50.5%
Age 65+	28.5%	36.5%
Held by households 46+	71.8%	87.0%
Held by households with no children at home	54.9%	72.5%

The concentration is in the data, not only in the model. In the ACS, 72 percent of the family-sized owner stock is held by post-fertile households and 55 percent by households with no children at home. The model has the same ordering and overstates the concentration among older households. Its model column will be recomputed after the timing correction. The data column already shows the allocation problem that motivates the model.

A preliminary retention experiment moves prices by less than one percent and completed fertility by about one hundredth. The result mainly reveals that the current old households are close to a retain-everything corner: without a health, mortality, or downsizing margin, few of them respond to a disposal wedge. I treat this exercise as a runtime pattern, not as an estimate of the old side of the allocation wedge.

8 Preliminary Policy Patterns

The instruments follow the access margins of the compact model: a holding tax on large owner units (the capitalization channel of eq. 30), birth-linked down-payment grants restricted to family-sized rungs (the financing gap $\zeta^{O,F}$), variation in the rental cap $\bar{h}^R$, a retention wedge on old owners, and a revenue-linked package combining the tax and the grant. These exercises describe the July diagnostic runtime, not final policy estimates. The grant enters both household values and the forward distribution in that runtime. Prices re-clear in every fixed-population experiment, but all cases will be rerun after re-estimation.

The current tax experiment produces capitalization and a small birth response. A two percent holding tax on large units lowers their asset price by 12 percent. Capitalization is similar across the grids, moment weights, and owner menus that have been checked, while completed fertility rises by +0.004 to +0.009. As eq. 30 predicts, down payments fall. In this snapshot, exposed households buy near the fertility deadline with wealth already above the threshold, so cheaper access changes which house they buy more than whether they have children.

The grant runtime produces a dose–response. A down-payment grant paid at a birth, restricted to family-sized rungs, raises completed fertility by +0.01, +0.13, +0.26, and +0.49 at grant sizes 0.1, 0.4, 0.58, and 1.0, in units of mean annual income. In the current solution the access instrument has a larger response because it is conditional on the birth itself: it closes the financing gap at the moment the household is on its fertility margin. The conditioning also distorts. Small grants delay pre-birth purchases, since households near a birth wait for the subsidy, and young ownership falls from 0.23 to 0.09 at the smallest grant. The distortion fades at larger grant sizes, but any conditional instrument inherits the margin.

The combined experiment follows both sides of Section 3's reallocation. The two percent tax on large homes funds the birth-linked grant; the tax lowers incumbents' retention value and the price of large units by 11 percent, and the grant closes the young side's financing gap at the birth margin. At the July calibration the package raises completed fertility by +0.15 and is revenue-funded. This magnitude is provisional.

8.1 The entry and scale margin

The results above hold population scale fixed at its benchmark normalization, $S = 1$, to isolate changes in household behavior. The full equilibrium of Section 4.6 also imposes the entry and scale condition. In those results, a policy changes household values, the probability of entering against the outside option, and the number of locally born future entrants. These changes determine S through equation (27), and the housing price re-clears using level demand in equation (29). The outside value and entrant pool are recovered at the benchmark so that $S = 1$, then held fixed across counterfactuals. The maintained demographic assumption is that locally born children become future potential entrants one-for-one.

These results use $N_b = 120$ at the July calibration. The fixed-population column re-solves each case inside the same protocol. The grant and package rows carry a few-percent price imprecision from the nested scale-price solver's fast iterations.

Table 5: July 9 policy runtime with the entry margin on ($N_b = 120$; provisional)

Policy	ΔTFR, fixed pop.	ΔTFR, intensive	Scale S	Total births
Holding tax 2%	+0.004	+0.002	1.003	+0.4%
Grant 0.4, rungs ≥ 6	+0.119	+0.083	1.038	+8.2%
Package (tax + grant)	+0.162	+0.086	1.040	+8.6%
Rental cap 6.5	−0.006	−0.009	1.004	$\approx$ 0.0%
Retention wedge 0.2	+0.006	+0.000	1.007	+0.7%

In the current runtime, the grant raises population scale and the resulting price response offsets part of its fixed-population fertility effect. Total births nevertheless rise in the grant and package rows. The holding tax and retention wedge remain close to neutral. These patterns distinguish per-family fertility from births generated in the modeled market. Whether a local birth gain represents creation or relocation remains outside the one-market model.

9 Current Repairs and Re-estimation

The first repair concerns timing. The Bellman problem gives a household its new tenure and house in the current period, but the July distribution code uses inherited tenure for market clearing and several moments. The repaired code evaluates current housing demand and moments using the current tenure and house probabilities, while the same choices enter the transition to the next-period state. It also includes a same-period purchase in the housing response measured at a first birth. The fixed-parameter comparison shows that this correction is quantitatively important, so the previous parameter vector must be re-estimated.

A separate problem was reproducibility. The July Torch runtime used a normalized working-age income profile, including the intended low-income entry cell, and a dense wealth grid on $[-12, 30]$.

Those settings were absent from the production copy. They have now been restored in a named configuration together with the five owner rungs, target definitions, and search and verification grids. The grant forward map has also been ported to the main package, and only solutions meeting the strict market-clearing tolerance are eligible in the new search. The remaining sequence is to complete the $N_b = 120$ re-estimation, verify the winner at $N_b = 240$, regenerate the diagnostic packet, and rerun the policy table.

The paper-side specification is already aligned with the estimated object in three places. Owner utility applies χ_O to usable space $H_k - \bar{h}(n, s)$, bequest utility retains its family-size-dependent level, and all owners face the same debt floor. Those changes describe the existing estimated preferences and finance rule; they are not new mechanisms.

10 Limitations

The model remains deliberately incomplete after these repairs. Old-age ownership is too high because health, mortality, and other reasons for downsizing are absent. Fertility has no age-dependent fecundity, so waiting is costless until the final fertile period. Both omissions matter for the timing and stock-allocation questions at the center of the paper.

The computed benchmark prices old-owner retention through transaction costs only. The capital-gains and basis mechanism of Section 3 is not in the quantitative state space. The supply curve is unit-elastic, while the compact misallocation result holds the stock fixed. Finally, the entry exercise relies on an externally recovered outside value and on the convention that local births become future entrants. The entry table is therefore a decomposition, not an estimated migration response.

11 Conclusion

Children use housing space, and access to that space differs sharply by tenure. A rental-size limit raises the shadow price of room for a constrained household. Ownership relaxes the size limit but introduces a down payment. Retention frictions can keep large homes with older households whose current use of space is low. The compact model places these margins in one goods-equivalent access gap and shows how the same gap enters the private price of children.

The July quantitative snapshot suggests that this mechanism is active. Many households near the fertility margin value additional space above its user cost, purchases bunch late in the fertile window, and the data show that older households hold most family-sized owner housing. The size of these effects is not yet settled. Current tenure timing and the first earnings cell affect targeted moments and require a new calibration.

The immediate goal is therefore narrower than a new model extension: complete the timing-corrected re-estimation, verify it on the finer grid, and rerun the policy exercises from one recorded production configuration. The theory and the data fact organize that exercise. The corrected fit and policy table will determine how much quantitative weight the mechanism can carry.

A Additional Analytical Results

This appendix records three extensions to the short analytical note. They are not needed for the main argument in Section 3.

First, at equal scaling, $\chi = (1+k)\kappa q^m/\alpha$, the fixed-mode response to a marginal relaxation of a binding housing cap can be written as

$$\Psi_i^m = \frac{\kappa}{\alpha} \frac{\zeta_i^m}{c_i^m - \chi n_i^m}\, \omega_i^m, \qquad \omega_i^m > 0.$$

The normalized access gap therefore governs both the response of family size among incumbent households and the response of entry into the modeled market.

Second, for a collateral-capped owner, $H_i^O = a_i\rho/[(1-\phi)q]$, with $\rho = r + \delta + \tau^p$. Holding the user cost and the old-owner gains wedge fixed,

$$\left.\frac{\partial H_i^O}{\partial \tau^p}\right|_{q,\bar{\ell}} = \frac{a_i}{(1-\phi)q} > 0. \tag{30}$$

A property-tax increase can therefore relax the down-payment constraint by lowering the capitalized asset price, even when the housing user cost is held fixed.

Finally, a policy can move households across the rent–own boundary as well as relax a constraint within a tenure. The resulting change in births depends on both the mass of households near that boundary and the difference in family size between their owner and renter choices. Its sign is not determined by the direction of tenure switching alone.

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